# Nonlinear Practice 3

Module 8 - Nonlinear Pharmacokinetics · Daily practice sheet · No date

Name _____ Date _____

1. A 78-kg, 28 year old male is receiving phenytoin for the treatment of seizures. When this patient was taking a daily oral dose of 300 mg, his steady state plasma concentration was 10.4 mg/L. The dose was increased to 400 mg which resulted in a steady-state plasma concentration of 22 mg/L. Assume 100% bioavailability and an apparent volume of distribution that is equal to 70% of body weight.

**a** Calculate KM

**b** Calculate Vmax

**c** Calculate the half-life and clearance in this patient at steady-state while he is taking 300 mg/day. What is the half-life?

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**d** Calculate the half-life and clearance in this patient at steady-state while he is taking 400 mg/day. What is the half-life?

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**e** What is the daily phenytoin dose that should achieve a steady-state phenytoin plasma concentration of 15 mg/L in this patient? Please round to the nearest 10 mg.

# Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

## a

Her key

**Answer:** 9.4 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **R1** | 300 mg | the first daily dose |
| **$C_1$** | 10.4 mg/L | steady state on 300 mg |
| **R2** | 400 mg | the increased daily dose |
| **$C_2$** | 22 mg/L | steady state on 400 mg |
| **weight** | 78-kg | *not needed for KM* |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$$, with R1 = 300 mg/day, $C_1$ = 10.4 mg/L, R2 = 400 mg/day, $C_2$ = 22 mg/L

Two daily doses in the same patient and the steady-state concentration each produced: the direct method. The 78 kg weight and the 70% volume play no part in KM; they are needed only for the half-life in parts c and d.

ALGEBRA

$$KM = \frac{400 - 300}{(300/10.4) - (400/22)} = \frac{100}{28.846 - 18.182} = \frac{100}{10.664} = 9.377 \text{ mg/L}$$

Each dose over its own concentration, the earlier pair first so that the denominator is positive. The mg/day cancel top and bottom and mg/L moves to the top, so KM comes out as a concentration.

ROUND

KM = 9.4 mg/L

9.377 mg/L to one decimal place is 9.4 mg/L, her sheet's value, which her sheet carries into Vmax and every later part. The unrounded 9.377 mg/L is also accepted.

> **DOES THE NUMBER MAKE SENSE?**
>
> A one-third rise in dose more than doubled the concentration, from 10.4 to 22 mg/L, so elimination is saturable; KM is a concentration: 9.4 mg/L.

## b

Her key

**Answer:** 571 mg/day

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **KM** | 9.4 mg/L | from part a, added to $C_{ss}$ |
| **R1** | 300 mg | the dose used |
| **$C_1$** | 10.4 mg/L | its own steady state |

| | | |
|---|---|---|
| **R2, $C_2$** | 400 mg, 22 mg/L | the check pair |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Vmax} = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line solved for Vmax: multiply both sides by (KM + $C_{ss}$) and divide by $C_{ss}$. With KM known from part a, one dose and the steady-state concentration it produced are enough.

ALGEBRA

$$\text{Vmax} = \frac{(300\ \text{mg/day})(9.4 + 10.4)\ \text{mg/L}}{10.4\ \text{mg/L}} = \frac{5940}{10.4} = 571.2\ \text{mg/day}$$

Her sheet uses the first pair, 300 mg/day with its own 10.4 mg/L. mg/L cancels and Vmax keeps the units of the dose, mg/day.

ALGEBRA

$$\text{Check: } \frac{(400\ \text{mg/day})(9.4 + 22)\ \text{mg/L}}{22\ \text{mg/L}} = \frac{12\ 560}{22} = 570.9\ \text{mg/day}$$

The second pair agrees within rounding, as it must for one patient with one Vmax. With KM unrounded, 9.377 mg/L, both pairs give 570.5 mg/day, inside the same tolerance.

ROUND

$$\text{Vmax} = 571\ \text{mg/day}$$

571.2 mg/day to the nearest milligram per day is her sheet's 571 mg/day, which her sheet carries into the half-life, clearance and dose of parts c to e.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax must exceed 400 mg/day, the higher dose, because that dose reached a steady state: 571 mg/day.

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**c** Her key

## **Answer:** 1.3 days

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **weight** | 78-kg | $V_D = 0.7 \times 78 = 54.6$ L |
| **$V_D$** | 70% of body weight | 0.7 L/kg |
| **$C_p$** | 10.4 mg/L | the steady state on 300 mg/day |
| **KM** | 9.4 mg/L | from part a |
| **Vmax** | 571 mg/day | from part b |

### THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.7\ \text{L/kg} \times 78\ \text{kg} = 54.6\ \text{L}$$

"70% of body weight" is read as 0.7 L/kg, taking 1 L as 1 kg, so the volume is 0.7 times the 78 kg weight, in litres, which the half-life line needs.

SETUP

$$t½ = V_D(\frac{0.693}{Vmax})(KM + C_p), \text{ at } C_p = 10.4 \text{ mg/L}$$

On 300 mg/day the patient is at the steady state that dose produced, 10.4 mg/L, and the half-life of a saturable drug is calculated at its concentration.

ALGEBRA

$$t½ = 54.6 \text{ L} \times \frac{0.693}{571 \text{ mg/day}} \times (9.4 + 10.4) \text{ mg/L} = 1.312 \text{ days}$$

L × mg/L gives mg, which cancels the mg of mg/day and leaves days. Leaving the volume as 0.7 gives 0.017, a result per kilogram, not a time.

ROUND

$$t½ = 1.3 \text{ days}$$

1.312 days to one decimal place, as her sheet gives it. On 400 mg/day the same line gives a longer half-life, because KM + $C_p$ is larger while Vmax stays fixed.

**DOES THE NUMBER MAKE SENSE?**

Vmax is in mg/day, so the half-life comes out in days: 1.312 days, more than one day and less than two.

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**c** Her key

**Answer:** 28.8 L/day

**WHAT THE STEM GIVES**

| | | |
|---|---|---|
| **$C_p$** | 10.4 mg/L | the steady state on 300 mg/day |
| **KM** | 9.4 mg/L | from part a |
| **Vmax** | 571 mg/day | from part b |
| **weight** | 78-kg | *not needed for clearance* |

**THE WORKING, ONE LINE AT A TIME**

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}, \text{ at } C_p = 10.4 \text{ mg/L}$$

On 300 mg/day the patient is at 10.4 mg/L, and the clearance of a saturable drug is calculated at its concentration, which sits in the denominator.

ALGEBRA

$$Cl_T = \frac{571 \text{ mg/day}}{(9.4 + 10.4) \text{ mg/L}} = \frac{571}{19.8} = 28.84 \text{ L/day}$$

mg/day over mg/L leaves L/day, a volume of plasma cleared per unit time. The weight and the volume of distribution are not needed for clearance.

ROUND

$$Cl_T = 28.8 \text{ L/day}$$

28.84 L/day to one decimal place, her sheet's value. On 400 mg/day the clearance is lower, because the larger $C_p$ makes the denominator larger.

**DOES THE NUMBER MAKE SENSE?**

KM + $C_p$, 19.8 mg/L, is about twice KM, so clearance is about half of Vmax ÷ KM: 28.84 L/day.

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**d** Her key

**Answer:** 2.1 days

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **weight** | 78-kg | $V_D = 0.7 \times 78 = 54.6$ L |
| **$V_D$** | 70% of body weight | 0.7 L/kg |
| **$C_p$** | 22 mg/L | the steady state on 400 mg/day |
| **KM** | 9.4 mg/L | from part a |
| **Vmax** | 571 mg/day | from part b |

### THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.7 \text{ L/kg} \times 78 \text{ kg} = 54.6 \text{ L}$$

"70% of body weight" is read as 0.7 L/kg, so the volume is 0.7 times the 78 kg weight, in litres. It does not change with the dose; only KM + $C_p$ does.

SETUP

$$t½ = V_D\left(\frac{0.693}{Vmax}\right)(KM + C_p), \text{ at } C_p = 22 \text{ mg/L}$$

On 400 mg/day the patient is at the steady state that dose produced, 22 mg/L, so the half-life line is evaluated at that concentration.

ALGEBRA

$$t½ = 54.6 \text{ L} \times \frac{0.693}{571 \text{ mg/day}} \times (9.4 + 22) \text{ mg/L} = 2.081 \text{ days}$$

The same line as in part c with $C_p$ = 22 mg/L. L × mg/L gives mg, which cancels against mg/day and leaves days.

ROUND

$$t½ = 2.1 \text{ days}$$

2.081 days to one decimal place, her sheet's value. A one-third rise in dose took the half-life from 1.3 to 2.1 days, because $C_p$ rose from 10.4 to 22 mg/L.

**DOES THE NUMBER MAKE SENSE?**

The half-life at 22 mg/L must be longer than at 10.4 mg/L, because KM + $C_p$ is larger: 2.081 days.

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**d** Her key

**Answer:** 18.2 L/day

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$C_p$** | 22 mg/L | the steady state on 400 mg/day |
| **KM** | 9.4 mg/L | from part a |
| **Vmax** | 571 mg/day | from part b |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}, \text{ at } C_p = 22 \text{ mg/L}$$

The same clearance line as in part c, now at the 22 mg/L that 400 mg/day produced, because the clearance of a saturable drug changes with concentration.

ALGEBRA

$$Cl_T = \frac{571 \text{ mg/day}}{(9.4 + 22) \text{ mg/L}} = \frac{571}{31.4} = 18.18 \text{ L/day}$$

mg/day over mg/L leaves L/day. The larger concentration makes the denominator larger, so the clearance is smaller than on 300 mg/day.

ROUND

$$Cl_T = 18.2 \text{ L/day}$$

18.18 L/day to one decimal place, her sheet's value. Clearance falls and half-life rises together as the dose rises, because both depend on KM + $C_p$.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance at 22 mg/L must be below its value at 10.4 mg/L, because $C_p$ is in the denominator: 18.18 L/day.

## e

Her key

**Answer:** 350 mg/day

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **KM** | 9.4 mg/L | from part a |
| **Vmax** | 571 mg/day | from part b |
| **$C_{ss}$** | 15 mg/L | the target |
| **rounding** | nearest 10 mg | 351.02 becomes 350 |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{Vmax \times C_{ss}}{KM + C_{ss}}$$

The patient's own KM and Vmax, from parts a and b, with the target of 15 mg/L as $C_{ss}$, in the nonlinear dose-rate line. Vmax is in mg/day, so no weight is needed.

ALGEBRA

$$R = \frac{(571 \text{ mg/day})(15 \text{ mg/L})}{(15 + 9.4) \text{ mg/L}} = \frac{8565}{24.4} = 351.02 \text{ mg/day}$$

mg/L cancels and leaves mg/day. 15 mg/L lies between the 10.4 and 22 mg/L measured, so the dose lies between the 300 and 400 mg/day he took.

ROUND

R = 350 mg/day

The stem asks for the nearest 10 mg, so 351.02 mg/day is reported as 350 mg/day, as her sheet gives it. 350 mg/day is a dose that can be given and holds $C_{SS}$ close to 15 mg/L.

**DOES THE NUMBER MAKE SENSE?**

15 mg/L lies between 10.4 and 22 mg/L, so the dose lies between 300 and 400 mg/day: 350 mg/day.